Please read the instructions below before the exam starts, and listen carefully to any instructions given by the invigilators. You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2023 **EC113 2022/3 A 800**

Only Casio FX-83GTPLUS or FX-85TGPLUS or FX-83GTx or FX-85GTx Casio FX- 85MS calculators are permitted
Two answer books

UNIVERSITY OF READING

Introductory Microeconomics (EC113)

Three hours

Answer **ALL** questions from Section A (32 marks), answer **TWO** questions from Section B (34 marks), and answer **TWO** questions from Section C (34 marks).

Section A

Answer ALL questions

1. Define opportunity cost and provide an example.

(2 marks)

2. Jo is faced with the following costs and benefits for activity X. What is the optimal level of activity X for Jo? Explain your answer.

(3 marks)

Units of Activity X	Total Cost	Total Benefit
1	20	200
2	50	300
3	90	380
4	140	430
5	200	450

3. Weakened economic growth in China and other emerging economies has led to a decrease in the global demand for copper. At the same time, the costs of extracting copper have increased, due to higher energy prices. What would we expect to happen to the equilibrium price and quantity of copper as a result?

(3 marks)

4. A consumer is currently consuming a bundle of good X and good Y. Given this consumption bundle, the consumer's marginal rate of substitution, MRS_{XY}, is less than the ratio of the prices of the two goods, P_X / P_Y. Is the consumer maximizing their utility? If not, how should the consumer change their consumption of the two goods in order to maximize their utility?

(3 marks)

5. If an increase in the price of coffee causes an increase in the total revenue of coffee shops, what can we determine about the price elasticity of demand for coffee? Justify your answer.

(3 marks)

6. Explain the difference between an income consumption curve and an Engel curve.

(3 marks)

7. Explain the law of diminishing returns. Does it always hold in the short run?

(3 marks)

8. The total cost (TC) of producing Nokia Express Music phones (Q) is given as: TC = 200 + 5Q. What is the average fixed cost, Marginal cost, and average variable cost?

(3 marks)

9. List the main characteristics of a competitive market?

(3 marks)

10. Draw the demand, marginal revenue, average total cost and marginal costs for a monopolist. Show the profit maximizing level of output, profit maximizing price and the amount of profit.

(3 marks)

11. Name one type of market failure and explain why it may cause market outcomes to be inefficient.

(3 marks)

Section B

Answer TWO questions

12. The market demand (Q_D) and supply (Q_S) for good X are given by:

$Q_D = 280 - 4P$

$Q_s = 240 + P$

(a) Calculate the equilibrium price and quantity and illustrate this equilibrium in a diagram.

(6 marks)

(b) What is consumer surplus? Calculate the value of the consumer surplus.

(3 marks)

(c) Calculate the price elasticity of demand at the market equilibrium. Is demand elastic or inelastic?

(3 marks)

(d) What is the difference between the own-price and the cross-price elasticity of demand? Explain two factors that affect the own-price elasticity of demand.

(5 marks)

13. Suppose a consumer has an income of $M = £450$ per week to spend on food and housing. The price of food is initially $P_F = £5$ / kg and the price of housing is $P_S = £10$ /sq m.

(a) In a diagram, with food on the horizontal axis and housing on the vertical axis, illustrate this consumer's budget constraint.

(2 marks)

(b) Illustrate and explain the impact on the consumer's budget constraint of: (i) an increase in the price of food to £15 / kg; (ii) a decrease in the consumer's income to £300 per week.

(3 marks)

(c) What do we mean by the marginal rate of substitution? What would be the value of the marginal rate of substitution if two goods are perfect (1 to 1) substitutes? (3 marks)

(d) Assume food is a normal good. Explain the direction of the substitution, income and total effects, if the price of food increases. You do not need to illustrate these, just explain the effects in words. (5 marks)

(e) Provide two examples of cognitive biases. Do they imply that individual decisions are not rational? (4 marks)

14. The market demand (Q_D) and supply (Q_S) for good X are:

$Q_D = 24 - P$ and $Q_S = 2P$

where Q_D and Q_S measure the quantity demanded and supplied, and P is the price in pounds.

(a) Calculate the equilibrium price and quantity.

(2 marks)

(b) Suppose the government levies a tax of £9 per unit on the sellers of good X. Calculate the new equilibrium price and quantity. How much of the tax is paid by the sellers and how much is paid by the buyers?

(6 marks)

(c) Draw a diagram to illustrate the impact of the tax. The diagram should show the initial equilibrium identified in a), the new equilibrium identified in b), the price paid by buyers and the price received by sellers.

(3 marks)

(d) Would your answer in b) change if instead the tax were levied on buyers? Why or why not?

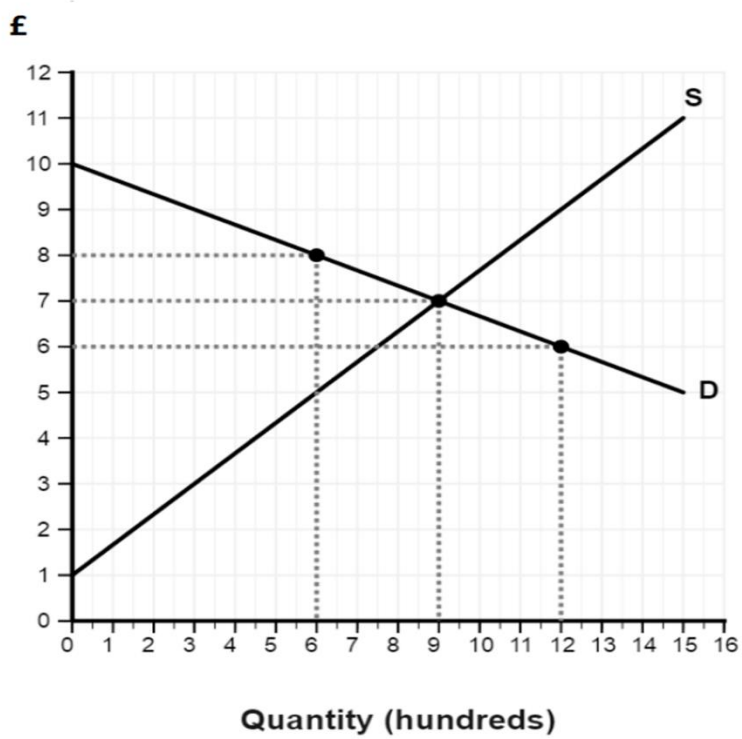
(3 marks)

(e) Will the tax be more effective if demand is elastic or inelastic? Explain.

(3 marks)

Section C
Answer TWO questions

15. The diagram below shows the market for taxi rides in Reading, UK.



(a) Suppose the Reading Borough Council sets a price ceiling of £5 for taxi rides. Complete the following table:

	Without price ceiling	With the price ceiling
Consumer surplus		
Producer surplus		
Total surplus		

(7 marks)

(b) Suppose the Council decides not to impose a price ceiling or to limit the number of taxi rides, but instead passes a law that says the price of a taxi ride cannot fall below £6. Is the new equilibrium price in this market higher than, less than, or equal to £7? Present and discuss a supply and demand diagram to explain your answer.

(5 marks)

(c) Now suppose the Reading Borough Council decides not to impose any price controls but Uber decreases their prices in Reading. Uber and taxis are substitutes for consumers. Is the new equilibrium price in this market higher than, less than, or equal to £7? Present and discuss a supply and demand diagram to explain your answer.

(5 marks)

16. British Box Company (BBC) produces cardboard boxes that are sold in bundles of 5000 boxes. The market is highly competitive, with boxes currently selling for £150 per bundle. BBC's total and marginal cost curves are:

$$TC = 1{,}500{,}000 + 0.002Q^2$$
$$MC = 0.004Q$$

where Q is measured in thousand box bundles per year.

(a) Calculate BBC's profit maximizing quantity. Is the firm earning a profit?

(5 marks)

(b) Analyse BBC's position in terms of the shutdown condition. Should BBC operate or shut down in the short run?

(5 marks)

(c) Using diagrams, explain the difference between long-run and short-run equilibrium in a perfectly competitive market.

(7 marks)

17. Rishi “the Dishi” Sunak is the city planner in a large city in Southeast England. The city is considering a proposal to award an exclusive contract to Kwarteng, Inc., a cable television carrier. Mr. Sunak has discovered that an economist hired a year before has generated the demand, marginal revenue, total cost and marginal cost functions given below:

$$Q = 160 - 4p$$
$$TR = 40Q - 0.25Q^2$$
$$MR = 40 - 0.5Q$$
$$TC = 4Q$$
$$MC = 4$$

where Q = the number of cable subscribers and P = the price of basic monthly cable service. Conditions change very slowly in the community so that Mr. Sunak considers the cost and demand functions to be reasonably valid for present conditions. Mr. Sunak knows relatively little economics and has hired you to answer the questions listed below.

(a) What price and quantity would be expected if the firm is allowed to operate completely unregulated?

(5 marks)

(b) Mr. Sunak has asked you to recommend a price and quantity that would be socially efficient. Recommend a price and quantity to Mr. Sunak using economic theory to justify your answer.

(7 marks)

(c) Compare the economic efficiency implications of (a) and (b) above. Your answer need not include numerical calculations, but should include relevant diagrams to demonstrate the concept of deadweight loss.

(5 marks)

(End of question paper)